

INTERNATIONAL STANDARD

Specifications for particular types of winding wires – Part 32: Glass fibre wound, resin or varnish impregnated, bare or enamelled rectangular copper wire, temperature index 155



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IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland

Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00
info@iec.ch
www.iec.ch

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**Specifications for particular types of winding wires –
Part 32: Glass fibre wound, resin or varnish impregnated, bare or enamelled
rectangular copper wire, temperature index 155**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

SPECIFICATIONS FOR PARTICULAR TYPES OF WINDING WIRES –**Part 32: Glass fibre wound, resin or varnish impregnated,
bare or enamelled rectangular copper wire, temperature index 155**

FOREWORD

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International Standard IEC 60317-32 has been prepared by IEC technical committee 55: Winding wires.

This second edition cancels and replaces the first edition published in 1990, Amendment 1:1997 and Amendment 2:2005. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- new 3.2.2 containing general notes on winding wire, formerly part of the scope;
- introduction of glass fibre coverings over grade 1 enamelled conductor in 3.2.2
- revision to references to IEC 60317-0-4 to make clear that their application is normative

The text of this standard is based on the following documents:

CDV	Report on voting
55/1492/CDV	55/1515/RVC

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

This International Standard is to be used in conjunction with the IEC 60317-0-4:2015.

The numbering of clauses in this standard is not continuous from Clauses 20 and 30 in order to reserve space for possible future wire requirements prior to those for wire packaging.

A list of all the parts in the IEC 60317 series, published under the general title *Specifications for particular types of winding wires* can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

This part of IEC 60317 is one of a series which deals with insulated wires used for windings in electrical equipment. The series has three groups describing:

- 1) Winding wires – Test methods (IEC 60851);
- 2) Specifications for particular types of winding wires (IEC 60317);
- 3) Packaging of winding wires (IEC 60264).

SPECIFICATIONS FOR PARTICULAR TYPES OF WINDING WIRES –

Part 32: Glass fibre wound, resin or varnish impregnated, bare or enamelled rectangular copper wire, temperature index 155

1 Scope

This part of IEC 60317 specifies the requirements of glass fibre wound, resin or varnish impregnated, bare or enamelled rectangular copper winding wire, temperature index 155.

NOTE For this type of wire, the heat shock test is inappropriate and therefore a heat shock temperature cannot be established. Consequently, a class based on the requirements for temperature index and heat shock temperature cannot be specified.

The range of nominal conductor dimensions covered by this standard is:

- width: min. 2,0 mm max. 16,0 mm;
- thickness: min. 0,80 mm max. 5,60 mm.

The specified combinations of width and thickness as well as the specified width/thickness ratio are according to IEC 60317-0-4.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60317-0-4:2015, *Specifications for particular types of winding wires – Part 0: General requirements – Section 4: Glass fibre wound, resin or varnish impregnated, bare or enamelled rectangular copper wire*

3 Terms, definitions, general notes and appearance

3.1 Terms and definitions

For the purposes of this document, the terms and definitions given in 3.1 of IEC 60317-0-4:2015 apply.

3.2 General notes

3.2.1 Methods of test

Subclause 3.2 of IEC 60317-0-4:2015 applies.

In case of inconsistency between IEC 60317-0-4 and this standard, IEC 60317-32 shall prevail.

3.2.2 Winding wire

The enamelled wire shall have a temperature index of at least 155 and shall be agreed between purchaser and supplier.

The temperature index of the wire is dependent upon the type of impregnating agent used. The impregnating agent applied to the glass fibres shall have a minimum temperature index of 155.

The covering shall have one of the following grades of thickness:

- GL1: one glass fibre covering over a bare conductor;
- GL2: two glass fibre coverings over a bare conductor;
- grade 1 GL1: one glass fibre covering (GL1) over grade 1 enamelled conductor (grade 1)
- grade 1 GL2: two glass fibre coverings (GL2) over grade 1 enamelled conductor (grade 1)
- grade 2 GL1: one glass fibre covering (GL1) over grade 2 enamelled conductor (grade 2)
- grade 2 GL2: two glass fibre coverings (GL2) over grade 2 enamelled conductor (grade 2)

3.3 Appearance

Subclause 3.3 of IEC 60317-0-4:2015 applies.

4 Dimensions

Clause 4 of IEC 60317-0-4:2015 applies.

5 Electrical resistance

Clause 5 of IEC 60317-0-4:2015 applies.

6 Elongation

Clause 6 of IEC 60317-0-4:2015 applies.

7 Springiness

Clause 7 of IEC 60317-0-4:2015 applies.

8 Flexibility and adherence

Clause 8 of IEC 60317-0-4:2015 applies.

9 Heat shock

Test inappropriate.

10 Cut-through

Test inappropriate.

11 Resistance to abrasion

Test inappropriate.

12 Resistance to solvents

Test inappropriate.

13 Breakdown voltage

Clause 13 of IEC 60317-0-4:2015 applies.

14 Continuity of insulation

Test inappropriate.

15 Temperature index

Clause 15 of IEC 60317-0-4:2015 applies.

16 Resistance to refrigerants

Test inappropriate.

17 Solderability

Test inappropriate.

18 Heat or solvent bonding

Test inappropriate.

19 Dielectric dissipation factor

Test inappropriate.

20 Resistance to transformer oil

Test inappropriate.

23 Pin hole test

Test inappropriate.

30 Packaging

Clause 30 of IEC 60317-0-4:2015 applies.

Bibliography

IEC 60264 (all parts), *Packaging of winding wires*

IEC 60317 (all parts), *Specifications for particular types of winding wires*

IEC 60851 (all parts), *Winding wires – Test methods*

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch